

Edge-fog computing for low-latency remote patient monitoring in heart failure, enabling early detection of decompensation and reducing hospital readmissions.

Healthcare × Remote patient monitoring × Edge computing · OS009 v4 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
65 is the world moving	58 can we play, can we win	MEDIUM 0 named in the evidence	€506m partial evidence · per year	LATER research and standards activi...	L1 16 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying an edge-fog computing architecture for remote patient monitoring in heart failure, enabling early detection of decompensation and reducing hospital readmissions. It targets healthcare providers and hospitals managing chronic heart failure patients, and it exists now because recent research demonstrates the technical feasibility and clinical promise of low-latency, AI-enabled monitoring at the edge.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€821m €67.3m – €3.9bn	€506m €41.4m – €2.4bn	€14.2m €1.2m – €67.4m	partial
Observed: tendered contract value, annualised	€70.8m	€70.8m	€2.0m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Healthcare (EU27_2020)	46,762 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Edge computing	14.9 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC_PCPU	2025
Annual value of one engagement	€970k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-07-22
Size-weighted buyer base	5,698 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC_PCPU is used as a proxy for Edge computing: Cloud compute purchase as nearest series. Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 2 declared geographies are outside the European reference data (CA, US); the estimate covers the European footprint only. Sector adoption is read from Eurostat's all-activities aggregate for part of this vertical, which the enterprise ICT survey does not cover separately: PROXY — ICT survey excludes health activities; PROXY — residential care

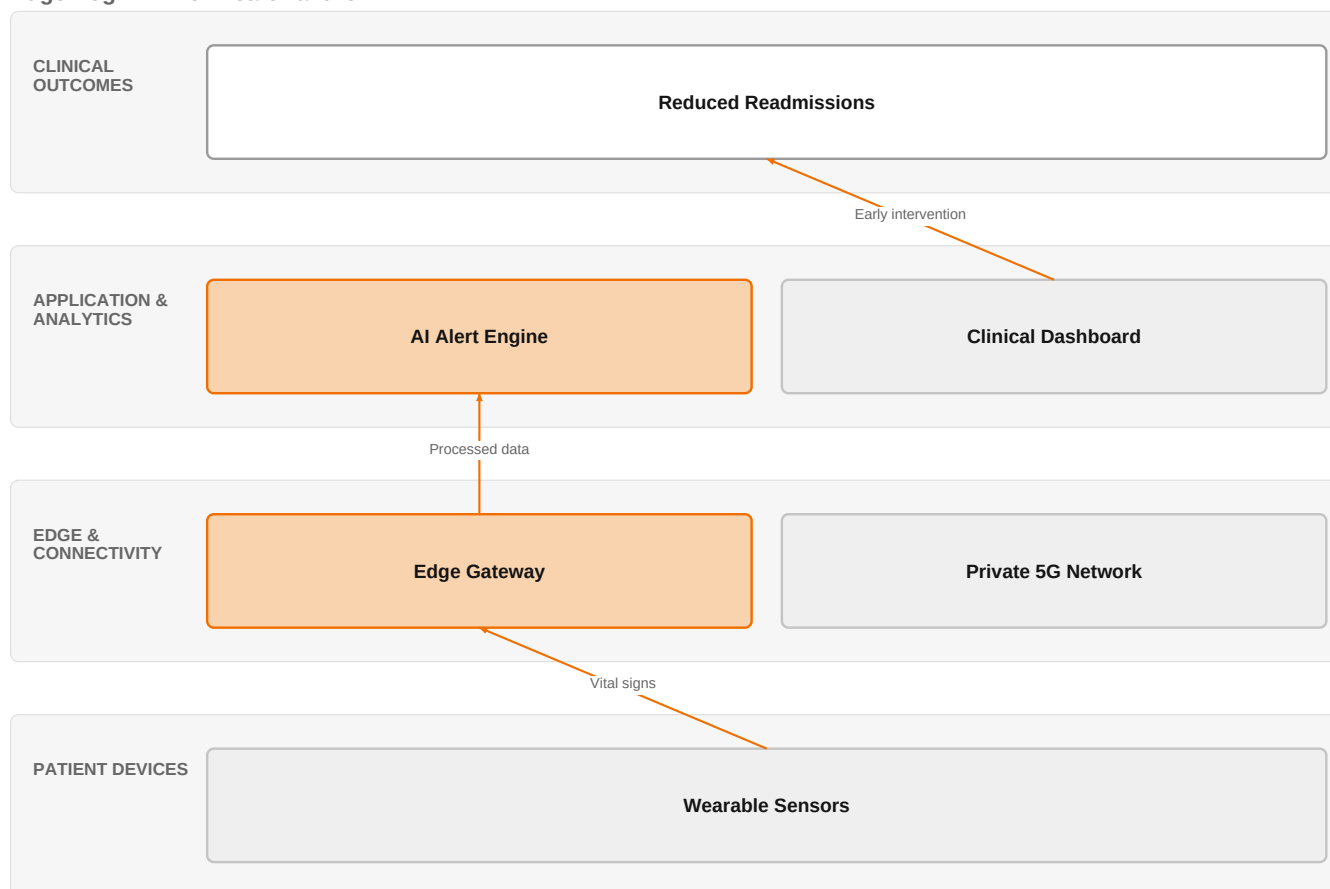
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€70.8m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-07-22
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 7 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 2 declared geographies are outside the European reference data (CA, US); the estimate covers the European footprint only.

The solution, in one picture

Edge-Fog RPM for Heart Failure



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how patient data flows from wearables through edge processing to clinical alerts, enabling early intervention and reduced readmissions.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Remote patient monitoring (RPM) has emerged as a key technological approach for chronic disease management, with sensor-integrated systems enabling real-time tracking of vital signs like heart rate, blood pressure, and oxygen saturation [SIG-C099858E391F]. In heart failure specifically, RPM using connected devices and telehealth platforms is seen as promising to detect early signs of decompensation [SIG-AFC3B44CAE18]. Recent technical advances include a NEWS2-based edge-fog architecture for continuous home care monitoring with low-latency clinical prioritization [SIG-08949F09B221], and a deep learning architecture optimized for edge deployment using 5G ultra-reliable low-latency communication for real-time vital sign monitoring and prediction [SIG-FF57F665F32F]. These developments shift the technical possibility from cloud-centric to edge-centric processing, enabling lower latency and more responsive clinical alerts.

Sources: SIG-C099858E391F Advances in Virus Dete 2026-06-24; SIG-AFC3B44CAE18 Global Cardiology Scie 2026-05-22; SIG-08949F09B221 Proceedings of the 16t 2026-01-01; SIG-FF57F665F32F openalex.org 2025-10-01

Who buys, and why now

The buyer is typically a chief medical officer or head of cardiology at a hospital or health system, while the clinical team (nurses, cardiologists) feels the pain of managing heart failure patients who are frequently readmitted. The trigger is often a high 30-day readmission rate or a mandate to reduce hospitalizations, prompting them to seek proactive monitoring solutions. They need to detect decompensation early to intervene before a crisis, but current monitoring often lacks the low-latency, continuous analysis required for timely alerts.

Why it is hot now — every claim bound to a dated source

- A deep learning architecture integrated with 5G enables real-time vital sign monitoring and prediction. [SIG-FF57F665F32F]
- A secure edge-cloud Internet of Medical Things platform integrates wearables and sensors for continuous monitoring of neurodegenerative diseases. [SIG-555CA62D7414]
- Traditional care models do not ensure continuous personalized monitoring for neurodegenerative diseases. [SIG-555CA62D7414]
- A secure edge–cloud Internet of Medical Things platform integrates wearables and sensors with cognitive and behavioral evaluations for neurodegenerative disease monitoring. [SIG-555CA62D7414]
- Continuous and personalized monitoring for neurodegenerative diseases is enabled by a secure edge-cloud IoMT platform integrating wearables and in-house sensors. [SIG-555CA62D7414]
- A deep learning architecture optimized for edge deployment using 5G ultra-reliable low-latency communication enables real-time vital sign monitoring and prediction. [SIG-FF57F665F32F]
- A NEWS2-based edge-fog architecture enables continuous home care monitoring with low-latency clinical prioritization. [SIG-08949F09B221]
- Remote patient monitoring using connected devices and telehealth platforms aims to detect early signs of decompensation. [SIG-AFC3B44CAE18]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble an end-to-end solution leveraging its IoT experts, Healthcare experts, and Orange Group researchers to design and deploy an edge-fog architecture. The engagement would start with a discovery phase using healthcare domain expertise, then deploy private mobile networks for reliable connectivity, integrate Enovacom / Enovalife for healthcare data interoperability, and ensure compliance with HDS certification for health data hosting. The solution would include edge computing nodes (potentially with partners like NVIDIA, Dell, or HPE) to run AI models locally, with a central cloud for management and analytics. The order would be: assess, design, deploy connectivity, deploy edge infrastructure, integrate with clinical systems, and operationalize with monitoring and support.

Why Orange can win

Orange can win by combining its healthcare-specific assets (Enovacom / Enovalife, HDS certification) with its connectivity expertise (private mobile networks) and research capabilities (Orange Group researchers). The HDS certification is a critical differentiator for health data hosting in France, and the integration of Enovacom / Enovalife provides a trusted bridge to hospital information systems. However, the edge computing hardware and AI models are not core Orange assets, so Orange would need to partner with NVIDIA, Dell, or HPE, which weakens the differentiation in the edge layer itself.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Private mobile networks	offer	L1	offer provides the technology in a matching domain	unconfirmed
Enovacom / Enovalife	offer	L1	offer addresses the use case but not this technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Dell	partner	L2	partner provides the required technology	unconfirmed
HPE	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
HDS (Hébergeur de Données de Santé)	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
MicroPort CardioFlow	reference	SUP	published reference in this vertical for this use case	unconfirmed
Bliksund	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Edge computing — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Edge computing; active in Healthcare; competes in Connectivity Solutions	structural	—
Lumen Technologies	Telecom operator peer	sells Edge computing; competes in Connectivity Solutions	structural	—
NTT DATA	Global systems integrator	sells Edge computing; competes in Connectivity Solutions	structural	—
Schneider Electric	Industrial / OT specialist	sells Edge computing	structural	—
HPE (Aruba, Athonet)	Network equipment vendor	sells Edge computing; competes in Connectivity Solutions — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Edge computing; competes in Connectivity Solutions	structural	—
Atos / Eviden	Global systems integrator	active in Healthcare	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current 30-day readmission rate for heart failure patients, and how are you tracking it?
- 2 Do you currently use any remote monitoring devices for heart failure, and if so, what are their limitations in terms of latency or alerting?
- 3 What is your existing IT infrastructure for handling patient data, and do you have any edge computing capabilities in place?
- 4 How do you currently manage data privacy and security for patient monitoring, and are you compliant with HDS or similar regulations?
- 5 What is your budget and timeline for implementing a new monitoring solution, and who would be the primary decision-maker?
- 6 Have you evaluated any cloud-based RPM solutions, and what were the main barriers to adoption?

Objections you will meet

They will say	You can answer
We already have a remote monitoring program using cloud-based platforms.	That's true, and many programs start there. However, cloud-based platforms often introduce latency that can delay critical alerts. An edge-fog architecture processes data locally, enabling near-real-time detection of decompensation, which is essential for early intervention.

The clinical evidence for RPM in heart failure is mixed.	You're right, the evidence is not conclusive across all trials. But recent studies show promise, especially with AI-enabled monitoring. Our approach focuses on low-latency edge processing, which addresses one of the key limitations in previous trials—timely data analysis.
We're concerned about the cost and complexity of deploying edge infrastructure.	Cost is a valid concern. However, by leveraging our private mobile networks and partnerships with hardware vendors, we can design a solution that fits your existing infrastructure and scales incrementally. We also ensure compliance with HDS, which can reduce regulatory overhead.

Risks and unknowns

The main risk is that the clinical evidence for RPM in heart failure is still uncertain—a systematic review and meta-analysis of 65 RCTs questions whether the cumulative evidence is robust [SIG-DBBEFE8CCC78]. Additionally, there is a 'reality check' on RPM adoption, suggesting barriers to real-world implementation [SIG-922D61D29366]. Unknowns include regulatory approval pathways for AI-based monitoring, reimbursement models, and whether hospitals will invest in edge infrastructure given budget constraints. The lifecycle state is 'fading,' indicating waning momentum.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European hospital group's remote patient monitoring tender, anchoring the solution on Enovacom / Enovalife, private mobile networks for reliable connectivity, and HDS certification, and include a technical appendix on the NEWS2-based edge-fog architecture for low-latency prioritization.
Sales	In a conversation with a healthcare provider's digital health lead, reference the MicroPort CardioFlow remote monitoring work and ask how their heart failure patients are currently monitored between visits, then introduce Enovacom / Enovalife as a bridge to continuous care.
Strategist / Innovator	Deep-dive into the NEWS2-based edge-fog architecture and the secure edge-cloud IoMT platform to assess technical feasibility and regulatory alignment with HDS and SecNumCloud for a healthcare-specific edge offering.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-C099858E391F	2026-06-24	Advances in Virus Detecti...	1	Remote Patient Monitoring
SIG-AFC3B44CAE18	2026-05-22	Global Cardiology Science...	1	Remote Patient Monitoring in Heart Failure
SIG-C6AE223DC81E	2026-04-23	ResearchHub Technologies,...	1	Advancements in Telemedicine and Remote Patient Monitoring
SIG-DBBEFE8CCC78	2026-02-27	openRxiv	1	Remote Patient Monitoring in Heart Failure: A Systematic Review, Meta-Analysis, and Trial Seque...
SIG-08949F09B221	2026-01-01	Proceedings of the 16th I...	1	NEWS2-Based Edge-Fog Architecture for Continuous Home Care Monitoring with Low-Latency Clinical...
SIG-555CA62D7414	2025-12-13	openalex.org	1	A Digital Health Platform for Remote and Multimodal Monitoring in Neurodegenerative Diseases
SIG-E95AED1CBD71	2025-12-10	openalex.org	1	Artificial intelligence-based remote monitoring for chronic heart failure: design and rationale...
SIG-4016692CAFB1	2025-10-21	Cureus	1	Reducing Emergency Department Strain Through Usable Wearables: A Proof-of-Concept for Multi-sen...
SIG-5A5443D644A0	2025-10-18	openalex.org	1	Invasive and Non-Invasive Remote Patient Monitoring Devices for Heart Failure: A Comparative Re...
SIG-FF57F665F32F	2025-10-01	openalex.org	1	Real-Time Health Monitoring Using 5G Networks: Deep Learning–Based Architecture for Remote Pati...
SIG-BE515E05C7A6	2025-08-29	EEPES 2025	1	A Prototype of Integrated Remote Patient Monitoring System
SIG-5EC51EEB6EC8	2025-08-21	openalex.org	1	From Clinic to Cloud: Efficacy of AIAssisted Remote Monitoring of Patients With Implantable Ca...
SIG-CD3D330696E2	2025-05-02	2025 International Confer...	1	IoT in Healthcare: Enhancing Remote Patient Monitoring
SIG-D583B366C8F3	2025-04-13	Deep Science Publishing	1	Artificial intelligence in patient monitoring: Enhancing care through real-time analytics and r...
SIG-077BFFEF1A9B	2025-04-07	Artificial Intelligence f...	1	Chapter 3 AI for remote patient monitoring in healthcare
SIG-922D61D29366	2026-05-26	bing.com	2	Remote patient monitoring faces a reality check

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:58+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.